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PALEOMAGNETIC EVIDENCE FOR TERTIARY ANTICLOCKWISE ROTATION IN SOUTHWEST PUERTO RICO

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Abstract. Late Cretaceous and Eocene paleomagnetic results from calcareous mudstone, limestone, and volcaniclastic rocks in southwest Puerto Rico yield a significant anticlockwise discordance in magnetization direction relative to North American reference data. Progressive thermal demagnetization has revealed stable, characteristic high blocking temperature magnetization components that display an intrasite bipolarity. The direction of this stable magnetization in tilt-corrected coordinates is consistently either northwest and down, or southeast and up. The pole position (39.4° N.Lat.; 209.6° E.Long.; dp = 9.3°; dm = 16.7°; N = 10) suggests post Eocene anticlockwise rotation of the sampling region relative to North America by 45° about a vertical axis. A rotating block domain model in which north-south compression is superimposed on WNW-ESE strike-slip faults in SW Puerto Rico can explain the inferred rotation.

Introduction

Studies from a variety of geologic disciplines indicate that the Northern Caribbean Plate Boundary is a complex tectonic feature characterized by strike-slip faults, pull-apart basins and paleo-subduction zones (Mann et al., 1984; Sykes et al., 1982). Recently, a "Puerto Rico Platelet" (McCann et al., 1987) has been defined based on earthquakes, microseismicity and marine seismic reflection records, suggesting Puerto Rico is a tectonically distinct block within the Northern Caribbean Plate Boundary.

Our objective is to use paleomagnetism in the structurally complex region of southwest Puerto Rico to determine the sense and magnitude of crustal/lithospheric block rotations relative to North America. Pre-Oligocene paleomagnetism is of particular interest because a major change in relative motions took place in the Eocene among the Caribbean, North American and South American plates (e.g. Malfait and Dinkelman, 1972; Pindell and Dewey, 1982; Burke et al., 1984; Duncan and Hargraves, 1984). This change heralded the prevailing easterly motion of Caribbean lithosphere relative to the North and South American plates and introduced strike-slip deformation to the Northern Caribbean Plate Boundary. In this paper we present new paleomagnetic data from southwest Puerto Rico, that provides paleolatitude information as well as a Late Cretaceous-Eocene pole position suitable for comparison to North American paleomagnetic reference data.

Geologic Setting and Field Sampling

We have sampled a variety of carbonate and volcaniclastic lithologies exposed in southwest Puerto Rico. These Late Cretaceous and Eocene units occur stratigraphically above the Bermeja Complex (the oldest known rocks on Puerto Rico) which comprises a linear belt of serpentinite, with rafts of Upper Jurassic to Lower Cretaceous amphibolites, spillites, cherts (Mariquita Chert) and the metavolcanic Cajul Basalts (Mattson, 1960; Mattson, 1973; Mattson and Pessagno, 1979; Volckmann, 1984; Schellekens et al., 1986). The sampled carbonate and volcaniclastic units are part of a larger, east-west trending central core of deformed volcanic, volcaniclastic and sedimentary rocks intruded by several plutons and overlain to the north and south by tilted Oligocene to Recent sedimentary rocks and sediments (see Figure 1).

Compressional deformation of this central core affected rocks of middle Eocene age (Glover and Mattson, 1960; Glover, 1971; Seiders et al., 1972; Mattson, 1973) producing southwest dipping thrust planes and tectonic transport towards the northeast (Glover and Mattson, 1960; Seiders et al., 1972; Mattson, 1973). The compression was followed by strike-slip and normal faulting (Glover and Mattson, 1960; Briggs and Pease, 1968; Glover, 1971; Mattson, 1973). Briggs and Pease (1968) suggest about 28 km of left lateral displacement along the southernmost Quebrada Vincente - Damian Arriba fault. Oligocene and younger sediments that overlie the WNW-ESE trending wrench faulting to the north appear to be unaffected, suggesting that this deformation style ended before the Oligocene. Geomorphology of east-west trending faults and the presence of recent seismicity indicates that deformation is still active today (Dart et al., 1980; Ascencio, 1980; McCann, 1985; Joyce et al., 1987).

Sampling site distribution is shown in Figure 1. In summary, we collected about seven samples per site, selecting

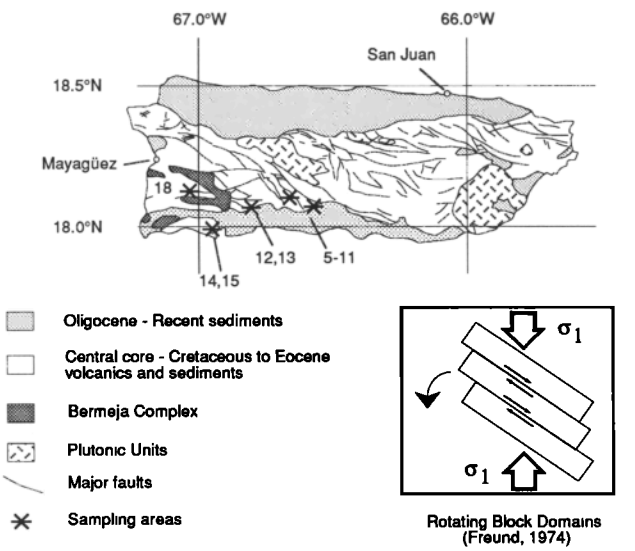


Fig. 1. A generalized geological map of Puerto Rico, redrafted from Almy (1969). Addition of major faults has been made using the map from Cox and Briggs (1973). The inset illustrates the rotating block domain model of Freund (1974).

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only the freshest appearing outcrops. Given the lack of good structural control in southwest Puerto Rico, we sampled only those exposures in which we could determine the "stratigraphic up" orientation, through criteria such as cross or graded bedding.

Yauco Formation (sites PR01, 02, 10, 12, 13, 17 and 18).
This formation is comprised of a variety of lithologic types described in detail by both Mattson (1960) and Pessagno (1960). The rocks are mainly mudstones and calcareous siltstones although there are occasional conglomerates, cherts and minor amounts of tuff. Based on foraminiferal studies, the total age range of the Yauco Formation is Cenomanian to early Maastrichtian (as reviewed by Volckmann, 1984).

Parguera Limestone (sites PR14-16).
The Parguera lithology is predominantly buff to light gray indurated limestone with tuff distributed at various locations in proportions of less than 10% (Mattson, 1960). The foraminifera of the Parguera limestone in southwest Puerto Rico indicate an age range of late Santonian to early Maastrichtian (Almy, 1965).

Monserate (sites PR03-09) and Anon formations (PR11).
The middle Eocene Monserate Formation varies in lithology from carbonate mudstones (sites PR03, 04) to volcaniclastic sandstones, siltstones and medium green vitric tuff (sites PR 05-09). A faint cross bedding can be observed in many of the volcaniclastic deposits. The Anon Formation lies conformably above and, in various locations interfingers with the Monserate unit (Mattson, 1966). The lithology of the Anon Formation comprises dacitic lavas in addition to the volcaniclastic siltstones we have sampled at site PR11. Planktonic foraminifera, distributed sporadically throughout these marine deposited siltstones have yielded an age of middle to late Eocene (Pessagno, 1960).

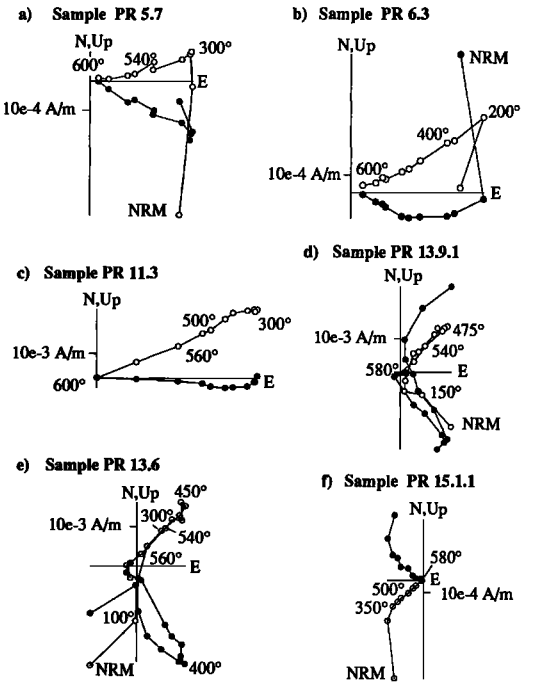


Fig. 2. Representative demagnetograms displaying progressive thermal demagnetization of southwest Puerto Rico samples. Applied temperatures are shown in °C and the directions here have been corrected for bedding tilt. Open (closed) symbols represent the vertical (horizontal) projection of the magnetization vector endpoint. Monserate Formation (a-b); Anon Formation (c); dual polarities in the Yauco Formation (d-e); and Parguera Limestone (f).

Paleomagnetism

All samples were measured using an A100 series United Scientific superconducting rock magnetometer (SRM) having sensitivity to magnetization intensity of 10^{-5} A/m. Our samples were stored in Mu-metal shielding compartments before and during laboratory analyses. Progressive alternating field (AF) and thermal demagnetization experiments were conducted using Schonstedt models GSD-1 single axis AC and TSD-1 thermal demagnetizers, respectively.

Progressive demagnetization
Approximately 20% of the samples were subjected to a detailed preliminary demagnetization study. This pilot suite of samples contained representatives from all 18 sites and all lithologies present in the collection. In this manner, optimum demagnetization schemes for various sites or lithologies were determined and utilized in subsequent analyses. All demagnetization work is expressed graphically using orthogonal vector diagrams (Zijderveld, 1967) through which components of magnetization can be visually isolated. Our data were subjected to computerized principal component analysis according to the method of Kirschvink (1980).

Representative thermal demagnetization diagrams are shown in Figure 2. Our detailed laboratory analyses revealed stable magnetizations in samples from 12 of the 18 sites and in general, the thermal demagnetization method was more successful than AF cleaning. All samples subjected to such thermal cleaning were near completely demagnetized by applied temperatures of 590-600°C. The measured stable components are of high unblocking temperature and directed characteristically along a northwest-southeast trend. While characteristic directions are present among the calcium carbonates, unstable magnetizations are exclusive to this lithology in the collection. Of the six thermally unstable sites, three (PR01, 02 and 10) are from the Yauco calcareous mudstone, two (PR03 and 04) are from the calcareous sandstones of the Monserate formation and the sixth site (PR16) is Parguera limestone. Attempts at AF demagnetization of samples from these sites were equally unsuccessful and were omitted from further analyses.

The orthogonal demagnetization diagrams of Figures 2a-c typify the response of volcaniclastic samples (from the Monserate and Anon formations) to thermal, step-wise cleaning. In each case beginning at about 150-200°C, a high-temperature magnetization component decays along a stable, origin-bound trajectory in response to increased ambient temperature. This component is directed characteristically toward the southeast with negative inclination.

Thermal demagnetization was successful in resolving both polarities of the characteristic magnetization from paleomagnetically stable Parguera Limestone and Yauco Mudstone, as shown in Figures 2d-f. Several individual samples from the Yauco Mudstone possessed both polarities of the characteristic magnetization, with the two antiparallel components having very little overlap in unblocking temperature spectra. For example, Figures 2d-e show the thermal demagnetization records of samples 13.06 and 13.09.1, respectively, in which initially a northwest and positive component is resolved between applied temperatures of 200 and 450°C. From this step onward to about 590°C, a characteristic component of opposite polarity (southeast and negative) decays along nearly an origin-bound trajectory. Figure 2f shows typical thermal demagnetization of Parguera Limestone. The resolved, high-temperature component is along the characteristic direction as well. Although these samples were among the weakest in our collection, stable components were still resolvable via the thermal method.

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Site mean directions.

Principal component analysis was used to examine the high unblocking temperature components which were combined into 12 site mean directions. These data along with corresponding Fisher (1953) statistics are listed in Table 1. With the exception of two sites, the mean directions are associated with very good statistical evaluations (i.e. low $\alpha 95$ values and, in general very high k values). The two "poor" sites, PR07 and PR17 have $\alpha 95$ values of 51.6° and 43.5°, respectively, and these have been excluded from the overall mean. In the case of site PR17, the large cone of confidence may be attributed to the small number of principal component directions used in the computation (N = 3, see Table 1). Mean data for site PR13 has the lowest $\alpha 95$ value (5.7°) and was calculated using 9 principal components: the most for any one of the 12 sites.

Site mean directions in both geographic and tilt-corrected coordinates are displayed in the stereographic projections of Figure 3. Among the site means a bipolar, northwesterly-southeasterly trend can easily be observed. Only one site mean (site PR14 of the Parguera limestone) may be considered amiss of this trend, having declination of 174.5° (see Figure 3 and Table 1). Although insignificant at the 95% level of confidence, the site mean directions are less dispersed following application of the bedding corrections. A more conclusive statement regarding the timing of remanence acquisition awaits a proper field test. For now, the presence of normal and reversed site mean directions, their improved grouping upon restoration of paleo-horizontal, and the fact that dissimilar lithologies bear similar high-temperature magnetization components all argue for a primary remanence.

Pole Position and Tectonic Interpretation

Ten of the twelve site mean directions listed in Table 1 were used in the calculation of a pole position for southwest Puerto Rico (sites PR07 and 17 have been excluded). The pole (as shown in Figure 4) is located at 39.4° North Latitude; 209.2° East Longitude (dp = 9.3°, dm = 16.7°) and the mean paleolatitude of our sampling region is 16°. Our pole is compared to the 50 Ma paleopole of Irving and Irving (1982) which is located at 82° N. Lat.; 189° E. Long., and by doing so we find that we can eliminate the discordance with a clockwise rotation

TABLE 1. Site mean data from SW Puerto Rico

Site	Rs	Uncorrected Dec.	Uncorrected Inc.	Corrected Dec.	Corrected Inc.	k	α95
PR05	6/6	112.2	-6.3	110.1	0.9	10.6	21.5
PR06	6/6	108.9	-23.8	116.0	-23.5	24.5	13.8
PR07	5/5	81.5	-5.9	92.7	31.1	3.2	51.6
PR08	8/8	130.6	-5.3	118.0	-54.4	5.9	24.9
PR09	6/6	123.2	0.1	113.2	-40.8	6.1	18.6
PR11	5/5	107.9	-30.3	100.7	-21.5	122.3	7.0
PR12	7/7	333.7	17.5	323.2	28.3	29.8	11.2
PR13	9/11	142.3	-14.6	143.7	-31.0	10.8	5.7
PR14	7/7	173.4	27.4	174.5	-19.5	6.6	16.5
PR15	5/5	327.6	16.9	312.6	40.6	113.2	7.2
PR17	3/3	291.0	-3.1	292.9	6.0	9.1	43.5
PR18	6/6	320.4	10.0	299.0	22.0	21.0	14.9
MEAN	10/12	313.8	10.5			10.1	16.0
"	"			306.9	29.8	11.1	15.1

Rs, ratio of number of samples used in computation to number of samples measured; k, precision parameter; $\alpha 95$, radius in degrees of the cone of 95% confidence; Dec., Inc., declination, inclination; Sites PR07 and PR17 eliminated from mean calculation due to high $\alpha 95$ values.

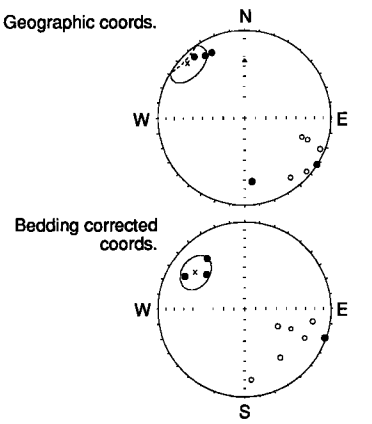


Fig. 3. Ten site mean directions plotted on equal angle stereographic projections in both geographic and bedding-corrected coordinates. Open (closed) symbols represent negative (positive) inclination. These data are listed in Table 1. The mean direction (x) of these ten site means is shown along with an ellipse of 95% confidence. The triangle shows the direction of the axially geocentric dipole field in southwest Puerto Rico.

of the sampling area by 45° about a vertical axis, thus suggesting anticlockwise rotation of these rock units since the Eocene.

Comparison of this southwest Puerto Rico pole with the North American apparent polar wander (NAAPW) path is an appropriate procedure, considering that we wish to understand the tectonic history of Puerto Rico relative to the North American plate, and that the Late Cretaceous to Recent interval of NAAPW is relatively well-known. As with other Caribbean paleomagnetic data, there is polarity uncertainty due to the low inclinations of the characteristic remanence. Age control is not sufficiently precise at any of the sites to permit unequivocal correlation to geomagnetic polarity time scales. However, given the paleolatitude of 16.0°, one might argue that the island could not have been as far south of the equator as 16° in the Late Cretaceous. It remains an assumption however, that the southwest Puerto Rico magnetizations were acquired in the northern hemisphere and that positive (negative) inclinations represent normal (reversed) directions. We have used the northwesterly and positive orientation of our data to calculate the pole position.

The detailed geometry of the thrust belt in SW Puerto Rico is not known. However, the general consistency of the site mean directions from different thrust sheets and different loca-

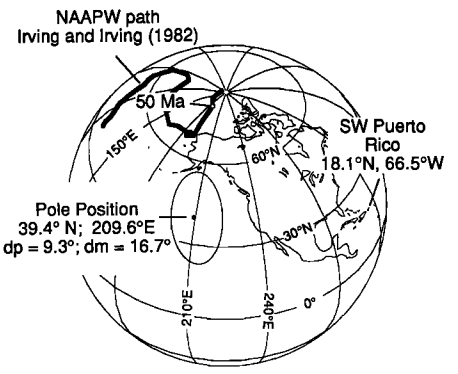


Fig. 4. A view of the Late Cretaceous-Eocene pole position calculated for southwest Puerto Rico. The ellipse of 95% confidence is also shown. When compared to Eocene reference data (the North American APW path of Irving and Irving, 1982), there is evidence for post-Eocene anticlockwise rotation of 45° about a vertical axis at the sampling site.

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tions within the stack, tends to suggest that the observed rotation occurred post- rather than syn- stacking. Although some variation in amount of rotation exists, it appears that southwest Puerto Rico has experienced a net anticlockwise rotation on a regional scale. We offer one possible scenario to explain the observed anticlockwise rotation in terms of the rotating block domain model of Freund (1974) in which the sense of rotation depends on the orientation of transcurrent faults relative to the axis of maximum horizontal compression. Crustal/lithospheric blocks and faults within a fault domain always rotate away from this axis. This model has been substantiated in a variety of tectonic environments with independently documented rotations (Scotti and Nur, 1987). McCann et al. (1987) demonstrate that today, the Puerto Rico platelet experiences compression at both its northern (Puerto Rico Trench) and southern (Muertos Trough) boundaries. Numerous WNW-ESE trending strike-slip faults exist in southwest Puerto Rico that are part of the Great Southern Puerto Rico faultzone. If the Puerto Rico platelet is under a state of north-south directed compression, then the WNW-ESE strike-slip faults in southwest Puerto Rico are of proper orientation to provide the anticlockwise sense of rotation (see Figure 1). The discordance may be associated with major compressive deformation that occurred between middle Eocene and early Oligocene time (Glover and Mattson, 1960, Seiders et al., 1972) which may be related to convergence of the North American and Caribbean plates.

References

Almy, C. C., Jr., Sedimentation and tectonism in the Upper Cretaceous Puerto Rican portion of the Caribbean island arc, *Trans. Gulf Coast Assoc. Geological Societies, 19,* 269-279, 1969.

Asencio, E., Western Puerto Rico seismicity, *U.S. Geol. Surv. Open-file Rept., 80-192,* 135 pp., 1980.

Briggs R. P., and M. H. Pease, Jr., Large and small scale wrench faulting in an island arc segment, Puerto Rico, *Abstracts for 1967, Geol Soc. Am. Special Paper, 115,* 24, 1968.

Burke, K., et al., Caribbean plate tectonics and relative plate motions, in *The Caribbean-South American Plate Boundary, Geol. Soc. Am. Mem.*vol.162, edited by W. E. Bonini, R. B. Hargraves, and R. Shagam, pp. 31-63, Boulder, Colo., 1984.

Cox, D. P., and R. P. Briggs, Metallogenic map of Puerto Rico, *U.S. Geol. Surv. Misc. Inv. Map I-721,* 1973.

Dart, R. L., et al., Puerto Rico seismic network data report of earthquakes located by the programs HYP071 and HYPOELLIPSE, July 1, 1975 - December 31, 1977, *U.S. Geol. Surv. Circ., 821,* 43 pp., 1980.

Duncan, R. A., and R. B. Hargraves, Plate tectonic evolution of the Caribbean region in the mantle reference frame, in *The Caribbean-South American Plate Boundary, Geol. Soc. Am. Mem.* vol. 162, edited by W. E. Bonini, R. B. Hargraves, and R. Shagam, pp. 81-93, Boulder, Colo., 1984.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London, Ser. A, 217,*295-305, 1953.

Freund, R., Kinematics of transform and transcurrent faults, *Tectonophysics, 21,* 93-134, 1974.

Glover L., III, Geology of the Coamo area, Puerto Rico, with comments on the Greater Antilles, *U.S. Geol Surv. Prof. Paper 636,* 102 pp., 1971.

Glover L., III, and P. H. Mattson, Successive thrust and transcurrent faulting during the early Tertiary in south-central Puerto Rico: in, Short papers in the geological sciences, *U.S. Geol. Surv., Prof. Paper 400-B,* 363-365, 1960.

Irving, E., and G. A. Irving, Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana, *Geophys. Surv., 5,* 141-188, 1982.

Joyce, J., et al., Onland active faulting in the Puerto Rico Platelet, *EOS Trans. A.G.U., 68,* 1483, 1987.

Kirschvink, J. L., The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc., 62,* 699-718, 1980.

McCann, W. R., On the earthquake hazards of Puerto Rico and the Virgin Islands, *Bull. Seism. Soc. Am.,* 75, 251-262,1985.

McCann, W. R., J. Joyce, and C. Lithgow, The Puerto Rico Platelet at the northeastern edge of the Caribbean Plate, *EOS Trans. A.G.U., 68,* 1483, 1987.

Malfait, B. I., and M. G. Dinkelman, Circum-Caribbean tectonic and igneous activity and the evolution of the Caribbean plate, *Geol. Soc. Am. Bull., 83,* 251-272, 1972.

Mann, P., et al., Neotectonics of Hispaniola: Plate motion, sedimentation, and seismicity at a restraining bend, *Earth Planet. Sci. Lett., 70,* 311-324, 1984.

Mattson, P. H., Geology of the Mayaguez area, Puerto Rico, *Geol. Soc. Am. Bull., 71,* 319-362, 1960.

Mattson, P. H., Geological characteristics of Puerto Rico: in, W.H. Poole (ed.), Continental Margins and Island Arcs, Papers of the Geological Survey of Canada, 66, 124-138, 1966.

Mattson, P. H., Puerto Rico - Virgin Islands: in, A. M. Spencer (ed.), Mesozoic-Cenozoic Orogenic Belts, Geological Society Spec. Pub. No. 4, London, 900 pp., 1973.

Mattson, P. H., and E. A. Pessagno, Jr., Jurassic and Early Cretaceous radiolarians in Puerto Rican ophiolite -- tectonic implications, *Geology, 440-444,* 1979.

Pessagno, E. A., Jr., Stratigraphy and micropaleontology of the Cretaceous and lower Tertiary of Puerto Rico, *Micropaleontology, 6,* 87-110, 1960.

Pindell, J., and J. F. Dewey, Permo-Triassic reconstruction of western Pangea and the evolution of the Gulf of Mexico-Caribbean region, *Tectonics, 1,* 179-211, 1982.

Schellekens, J. H., J. Joyce, and A. L. Smith, Origin and evolution of the Bermeja ophiolite complex, Puerto Rico, *EOS Trans. A.G.U., 67,* 1232, 1986.

Scotti, O., and A. Nur, Ball bearing vs. block rotation: evidence from crustal fault domains, *EOS Trans. A.G.U., 68,* 1483, 1987.

Seiders, V.M., R. P. Briggs, and L. Glover III, Geology of Isla Desecheo, Puerto Rico, with notes on the Great Southern Puerto Rico faultzone and the Quaternary stillstand of the sea, *U.S. Geol Surv. Prof. Paper 739,* 22 pp., 1972.

Sykes, L. R., et al., Motion of the Caribbean plate during the last 7 million years and implications for earlier Cenozoic movements, *J. Geophys. Res., 87,* 10,656-10,676, 1982.

Volckman, R. P., Upper Cretaceous stratigraphy of southwest Puerto Rico: a revision, *U. S. Geol. Surv. Bull., 1537-A, 73-83,* 1984.

Zijderveld, J. D. A., A.C. demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetism,* edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254-286, Elsevier Science, New York, 1967.

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